

Violet Ross

COMPUTER SCIENCE AT CU BOULDER

Boulder, CO

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Education

University of Colorado Boulder

PHD IN COMPUTER SCIENCE

- 3.94 GPA

Boulder, CO

Aug. 2025 - Present

Middlebury College

B.A. IN COMPUTER SCIENCE AND MATHEMATICS

- 3.96 GPA
- Phi Beta Kappa, Summa Cum Laude, Highest Honors Computer Science, High Honors Mathematics

Middlebury, VT

Sep. 2021 - May 2025

Experience

RESEARCH

University of Colorado Boulder

COMPUTER SCIENCE RESEARCH ASSISTANT

- Conducting mathematical methods research in network science and computational infectious disease epidemiology.

Boulder, CO

Aug. 2025 - Present

Middlebury College

COMPUTER SCIENCE RESEARCH ASSISTANT

- Designed a growth model for labelled hypergraphs and implemented it in Python.
- Utilized probabilistic and analytical methods to describe the relationship between parameter choice and the distribution of node labels.
- Verified these results by carrying out hypergraph growth simulations on Middlebury's high-performance computing cluster.

Middlebury, VT

Jun. 2024 - Aug. 2024

Yale University

MATHEMATICS RESEARCH ASSISTANT

- Participated in Yale University's "SUMRY" REU in mathematics on the Ambient Dimension of Manifold-fitting Data project.
- Implemented a model in R to non-trivially embed an intrinsically 3-dimensional manifold in any ambient dimension and then uniformly sample points from the embedding.
- Conducted research on the impact of high ambient dimension on the accuracy of intrinsic dimension estimators through both experiments and theoretical methods.

New Haven, CT

Jun. 2023 - Jul. 2023

Middlebury College

STATISTICS RESEARCH ASSISTANT

- Quantified social media responses to U.S. drone strikes in Yemen.
- Published a paper on these findings and gave talks at the 2023 Joint Mathematics Meetings and the 2023 Joint Statistical Meetings.

Middlebury, VT

Jun. 2022 - Aug. 2022

TEACHING

Middlebury College

MATH 0118: INTRODUCTION TO DATA SCIENCE TEACHING ASSISTANT

- Provided daily in-class programming and debugging support for students.
- Held two additional help sessions per week.
- Graded all homework assignments.

Middlebury, VT

6 Semesters

Middlebury College

CSCI 301: THEORY OF COMPUTATION TEACHING ASSISTANT

- Held two help sessions per week for students.

Middlebury, VT

1 Semester

Middlebury College

MATH 0218: STATISTICAL LEARNING TEACHING ASSISTANT

- Held one help session per week for students.

Middlebury, VT

1 Semester

Middlebury College

MATH 105: EXPONENTS, LOGARITHMS, AND FUNCTIONS (PRE-CALCULUS) TEACHING ASSISTANT

- Provided daily in-class problem-solving support to students.

Middlebury, VT

1 Semester

Publications

Investigating responses to US drone strikes in Yemen using Twitter data

Media, War, & Conflict

Evan Weiss, Violet Ross, Alex Lyford

2024

In Revision

Growing Hypergraphs with Homophily

arXiv

Violet Ross, Francis Cataldo, Philip S. Chodrow

2026

In Progress

Efficient Sampling for Modeling Viral Kinetics

Violet Ross, Daniel B. Larremore

2026

Talks

International School & Conference on Network Science

Boston, MA

“Modeling Homophilic Hypergraph Growth Using Edge Copying”

Jun. 2026

- Awarded Best Student Talk

Joint Mathematics Meetings

Boston, MA

“Quantifying Responses to U.S. Drone Strikes in Yemen Using Twitter Data”

Jan. 2023

Joint Statistical Meetings

Toronto, Canada

“Characterizing the Communities that Employ Academic Tracking Across the U.S.”

Aug. 2023

Middlebury College Introduction to Data Science Course

Middlebury, VT

Invited Guest Lecturer

Oct. 2022, Apr. 2023, Apr. 2024, Oct. 2024, Apr. 2025